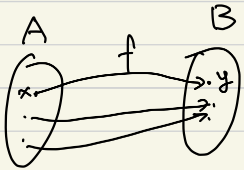
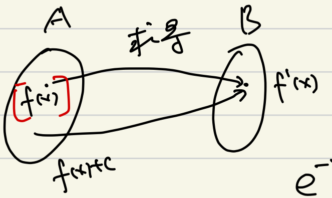
2022.11.7

Mon

Ch4 不定积分





e^{-x^2}

"求导"映射的逆映射. $\boxed{\text{不定积分}}$? N-L公式

本章: ① 定义 < 原函数存在性? 唯一性 (相差一个 C)

② 怎么求"原函数", 从 $f'(x)$ 求 $f(x)$

一般不好求

n个方法: < 查表 / 换元 / 分部积分

$[f(x)] = \{ f(x) + c \mid c \in \mathbb{R} \}$

· 变为单射, 注: $f_1'(x) = f_2'(x)$ 则 $f_1(x) = f_2(x) + C$

$+ \quad - \quad \times \quad \div \quad \circ$

§4.1. 不定积分的概念与性质

① 定义. 若可导函数 $F(x)$, $F'(x)=f(x)$, $x\in I$

则称 $F(x)$ 是 $f(x)$ 的一个原函数.（在区间 I 上）

eg: $\overset{f(x)}{=}\left(\frac{1}{x}\right)'$, $x\in(-\infty,0)\cup(0,+\infty)$, $F(x)=\frac{1}{x}+C$, $F(x)=\begin{cases}\frac{1}{x}+C_1, & x>0\\ \frac{1}{x}+C_2, & x<0\end{cases}$

"S"

记号: $\int \boxed{f(x)\,dx}$ 表示 $f(x)$ 的原函数(的集合)

不定积分

被积表达式

积分号 integrate

被积函数

积分变量

$\int f(x)$

$\int f$

$\int f\,dx$

例: $\int x\,dx = \frac{1}{2}x^2 + C$

☆定理: 原函数的存在定理. 即 $\int f(x)dx$ 是否有意义?

若 $f(x)$ 在 I 连续, 则存在 $F(x)$ 使 $F'(x)=f(x)$.

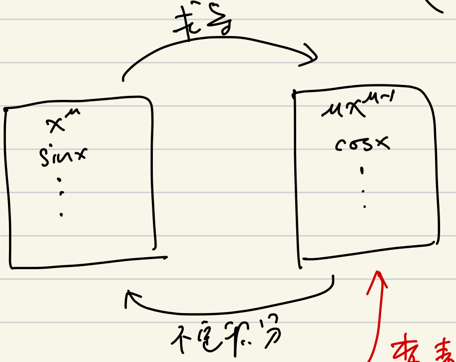
连续函数必存在原函数

why? 证明留到 ch5

2022.11.9

② 如何求不定积分

- 查表
- 积分换元
- 分部积分



$\int \boxed{f(x)}\,dx$ 查表

$(\ln x)' = \frac{1}{x}$

$(\ln(-x))'$

Tip：若 F_1, F_2 在表中，

$k_1F_1 + k_2F_2$ 可不放表里面

★ 定理（性质）"线性"

$$\int (k_1 f_1(x) + k_2 f_2(x))dx = k_1\int f(x)dx + k_2\int f_2(x)dx$$

(基本)积分表

1. $\int 0\mathrm{d}x = C.$
2. $\int 1\mathrm{d}x = \int \mathrm{d}x = x + C.$
3. $\int x^a\mathrm{d}x = \frac{x^{a+1}}{a+1} + C\ (a \neq -1, x > 0).$
4. $\int \frac{1}{x}\mathrm{d}x = \ln|x| + C$ [1] $(x \neq 0).$
5. $\int e^x\mathrm{d}x = e^x + C.$
6. $\int a^x\mathrm{d}x = \frac{a^x}{\ln a} + C\ (a > 0, a \neq 1).$
7. $\int \cos ax\mathrm{d}x = \frac{1}{a}\sin ax + C\ (a \neq 0).$
8. $\int \sin ax\mathrm{d}x = -\frac{1}{a}\cos ax + C\ (a \neq 0).$
9. $\int \underbrace{\sec^2 x}_{\to \frac{1}{\cos^2(x)}}\mathrm{d}x = \tan x + C.$
10. $\int \underbrace{\csc^2 x}_{\to \frac{1}{\sin^2(x)}}\mathrm{d}x = -\cot x + C.$
11. $\int \sec x \cdot \tan x\mathrm{d}x = \sec x + C.$ $= \frac{1}{\cos x}$
12. $\int \csc x \cdot \cot x\mathrm{d}x = -\csc x + C.$ $= \frac{1}{\sin x}$
13. $\int \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \arcsin x + C = -\arccos x + C_1.$
14. $\int \frac{\mathrm{d}x}{1+x^2} = \arctan x + C = -\operatorname{arccot} x + C_1.$

例: $\int \frac{\boxed{(x-1)^3}}{x^2} dx = \int \frac{x^3-3x^2+3x-1}{x^2} dx = \int x\,dx - 3\int dx + 3\int \frac{1}{x} dx - \int x^{-2} dx$

展开

$$= \frac{1}{2}x^2 - 3x + 3\ln|x| + \frac{1}{x} + C$$

例: $\int \tan^2 x\, dx = \int \frac{1}{\cos^2 x} dx - \int 1\, dx = \tan x - x + C$

$$\frac{\sin^2 x}{\cos^2 x} = \frac{1-\cos^2 x}{\cos^2 x}$$

例: $\int \frac{2x^4+x^2+3}{x^2+1} dx = \int \left(2\boxed{x^2}-\boxed{1}+\boxed{\frac{4}{x^2+1}}\right) dx = 2\,\frac{1}{3}x^3 - x + 4\arctan x + C$

$$\int f(x)\,dx = F(x) + C, \quad F'(x) = f(x), \quad dF(x) = f(x)\,dx$$

① $d\left(\int \boxed{f(x)\,dx}\right) = d(F(x)+C) = \boxed{f(x)\,dx}$

∴ "d" "$\int$" 可以抵消

记号和好处

② $\int dF(x) = \int f(x)\,dx = F(x) + C$ ☆

应用:

$$\left.\begin{aligned}\int f(x)\,dx &= F_1(x) + C_1 \\ &= F_2(x) + C_2\end{aligned}\right\} \Rightarrow F_1(x) = F_2(x) + C$$

得到一些恒等式.

2022.11.9
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§4.2 换元积分法 ， §4.3 分部积分法

求导 $\left\{\begin{array}{l} (k_1F_1(x)\pm k_2F_2(x))' = k_1F_1'(x)\pm k_2F_2'(x) \\ (F_1(x)F_2(x))' = F_1(x)F_2'(x)+F_1'(x)F_2(x) \\ (F_1(x)/F_2(x))' = \cdots \\ (F_1(\underbrace{F_2(x)}_{u}))' = F_1'(\overset{u}{F_2(x)})F_2'(x) \end{array}\right.$ $\xrightarrow[\text{积分}]{\text{不定}}$

① 线性性质
② 分部积分法
③ 换元法 $\left\{\begin{array}{l}\text{第Ⅰ类} \\ \text{第Ⅱ类}\end{array}\right.$

微分形式不变性

$$\int dF_1(F_2(x)) = \int dF_1(u) = \int F_1'(u)\,du = \int F_1'(F_2(x))F_2'(x)\,dx$$

☆ 换元法

Ⅰ: $\int \overset{\text{记为 } g(x)}{\boxed{f(\varphi(x))\varphi'(x)}}\,dx \xlongequal{u=\varphi(x)} \int f(u)\,du = F(u)+C = F(\varphi(x))+C$

Ⅱ: $\int f(u)\,du \xlongequal{u=\varphi(x)} \int \overset{\text{记为 } g(x)}{\boxed{f(\varphi(x))\varphi'(x)}}\,dx = G(x)+C = G(\varphi^{-1}(u))+C$

分部积分法：$\int u(x)v'(x)\,dx = \int u(x)\,dv(x) = u(x)v(x) - \int v(x)\,du(x)$

$= u(x)v(x) - \int u'(x)v(x)\,dx$

$$\int d(\underbrace{\boxed{u(x)v(x)}}_{u(x)v(x)+C}) = \int u(x)\,dv(x) + \int v(x)\,du(x)$$

知其一，则知另一个.

$$\int \cos(x)\,dx = \sin(x) + C$$
$$\int \cos(u)\,du = \sin(u) + C$$

例：$\int \boxed{2}\cos(\boxed{2x})\,d\boxed{x}$ $\quad \frac{\boxed{2x}}{2}$ u

$$= \frac{2}{2}\int \cos(u)\,du = \sin(u) + C = \sin(2x) + C$$

例：$\int \frac{1}{\underbrace{\boxed{3+2x}}_{=u}}\,dx$ $\quad x = \frac{u-3}{2}$, $dx = \frac{1}{2}du$

$$= \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln|3+2x| + C$$

例：$\int 2x e^{\boxed{x^2}=u}\,dx = \int e^{\boxed{x^2}}\,\boxed{2x\,dx}$ $\;= d(x^2)$ $\;= \int e^u\,du = e^u + C = e^{x^2} + C$

例：$\int \sin^3 x\,dx = \int \sin x \sin^2 x\,dx = \int \sin x(1-\cos^2 x)\,dx$

$$= \int \sin x\,dx - \int \underbrace{\cos^2 x}_{u^2}\,\boxed{\sin x\,dx}$$ $\quad d(\underbrace{-\cos x}_{u})$ $\quad , \int u^2\,du = \frac{1}{3}u^3 + C$

$$= -\cos x + \frac{1}{3}(+\cos x)^3 + C$$

第I类换元积分法：把表达式中某一部分看成整体！

例：$\int \sqrt{\boxed{1-x^2}}\,dx$ ，$x\in(-1,1)$

$1-x^2 = u^2$，$x=\sqrt{1-u^2}$，$x^2+u^2=1$

$x=\sin t$，$t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

$$\int\sqrt{1-\sin^2 t}\,d\sin t = \int \cancel{|}\cos t\cancel{|}\cos t\,dt = \int \cos^2 t\,dt$$

$$= \frac{t}{2}+\frac{\sin(2t)}{4}+C,\qquad \sin(2t)=2\sin(t)\cos(t)$$

$$= \frac{1}{2}\arcsin(x)+\frac{1}{2}x\sqrt{1-x^2}+C$$

例 14 求 $\int \cos^2 x\,dx$.

解 $\int \cos^2 x\,dx = \int \frac{1+\cos 2x}{2}dx = \frac{1}{2}\left(\int dx + \int \cos 2x\,dx\right)$

$= \frac{1}{2}\int dx + \frac{1}{4}\int \cos 2x\,d(2x) = \frac{x}{2}+\frac{\sin 2x}{4}+C.$

总结：第II类换元积分法，需要使用反函数

$$\int u v' dx = uv - \int u' v \, dx$$

$u(x)$ $v'(x)$

例. $\int \boxed{x e^x} \, dx = \int x (e^x)' \, dx = x e^x - \int 1 \cdot e^x \, dx = e^x (x-1) + C$

$\int \left(\frac{1}{2}x^2\right)' e^x \, dx = \frac{1}{2}x^2 e^x - \int \frac{1}{2}x^2 e^x \, dx$

$\int x e^x \, dx = \int x \, de^x = x e^x - \int e^x \, dx = e^x (x-1) + C$

例. "e^x" 看作 $d(e^x)$ "$\ln x$" 不看作 v'

$\frac{1}{x} dx$

$\int \underset{v'(x)}{\boxed{x}} \, \overset{u(x)}{\boxed{\ln x}} \, dx = \int \ln x \, d\left(\frac{1}{2}x^2\right) = \frac{1}{2}x^2 \ln x - \int \frac{1}{2}x^2 \, d\ln x$

$= \frac{1}{2}x^2 \ln x - \frac{1}{2}\int x \, dx = \frac{1}{2}x^2 \ln x - \frac{x^2}{4} + C$

例. 通过求导让别式变简单

例. $\int \ln x \, d\underset{v(x)}{x} = x \ln x - \int x \, \underset{\frac{1}{x}dx}{d\ln x} = x(\ln x - 1) + C$

比如: arcsin(x), arccos(x), arctan(x), … 都取作 u(x).

例：$\int \boxed{e^x} \sin x \, dx = \int \sin(x) \, de^x = e^x \sin(x) - \int e^x \cos(x) \, dx$

$\int e^x \cos(x) \, dx = \int \cos(x) \, de^x = e^x \cos(x) + \int e^x \sin(x) \, dx$

$\Downarrow$ 合并

$\int e^x \sin(x) \, dx = \frac{1}{2} e^x (\sin(x) - \cos(x)) + C$

注：$e^{i\theta} = \cos(\theta) + i \sin(\theta)$，$\int e^{(1+i)x} dx = \int e^{ax} dx = \frac{1}{a} e^{ax} + C$

$a = 1+i$

2022.11.11

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不定积分计算技巧

$$\int \frac{i\,du}{\sqrt{1-u^2}} = i \arcsin(u) + C$$

$$//$$

① $\int \sqrt{1-x^2}\,dx$. ② $\int \sqrt{1+x^2}\,dx$, ③ $\int \frac{dx}{\sqrt{1+x^2}}$, $x = iu$, ④ $\int \frac{dx}{\sqrt{x^2-1}}$, ⑤ $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C$ ⑥ $\int \sqrt{x^2-1}\,dx$

①
$$\begin{aligned}
\int \sqrt{1-x^2}\,dx &= x\sqrt{1-x^2} - \int x \, d\sqrt{1-x^2} , \\
&= x\sqrt{1-x^2} + \boxed{\int \frac{x^2}{\sqrt{1-x^2}}\,dx} \\
&= x\sqrt{1-x^2} + \boxed{\int \frac{x^2-1+1}{\sqrt{1-x^2}}\,dx} \\
&= x\sqrt{1-x^2} - \int \sqrt{1-x^2}\,dx + \int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C
\end{aligned}$$

$$\begin{aligned}
\left(\sqrt{1-x^2}\right)' &= \left((1-x^2)^{\frac{1}{2}}\right)' \\
&= \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot (-2x) \\
&= -\frac{x}{\sqrt{1-x^2}}
\end{aligned}$$

$$\Rightarrow \int \sqrt{1-x^2}\,dx = \frac{1}{2}\left(x\sqrt{1-x^2} + \arcsin(x)\right) + C$$

对此之后用换元法 令 $x = \sin t$

② $\int \sqrt{1+x^2}\,dx$, 方法1. $x = \tan t$. $1+x^2 = \frac{1}{\cos^2 t}$

方法2: $\int \sqrt{1+x^2}\,dx = \frac{1}{2}\left(x\sqrt{1+x^2} + \boxed{\int \frac{dx}{\sqrt{1+x^2}}}\right) + C$ ③

③ $\int \frac{dx}{\sqrt{1+x^2}} \overset{x=\tan t}{=\!=} \ln\left(x + \sqrt{x^2+1}\right) + C$

④ $\displaystyle\int \frac{dx}{\sqrt{x^2-1}} \xlongequal{x=\frac{1}{\cos t}} \ln\left|x+\sqrt{x^2-1}\right| + C$

分部积分

⑤ $\displaystyle\int \sqrt{x^2-1}\,dx = \frac{1}{2}\left(x\sqrt{x^2-1} - \ln\left|x+\sqrt{x^2-1}\right|\right) + C$

☆ 查表. or 软件 Mathematica.

☆ 验证

☆ 哪些不定积分是初等函数.

$\displaystyle\int \frac{P(x)}{Q(x)}\,dx$ (有理函数), $\displaystyle\int R(\sin(x), \cos(x))\,dx$, $\displaystyle\int R\left(x, \sqrt[n]{\frac{ax+b}{cx+d}}\right)dx$, $\displaystyle\int R\left(x, \sqrt{ax^2+bx+c}\right)dx$

$R(x) = \dfrac{P(x)}{Q(x)}$

$\sqrt[n]{\dfrac{ax+b}{cx+d}} = t$, $x = \dfrac{t^n d - b}{a - ct^n}$

配方

$1+t^2$, t^2-1, $1-t^2$

$t=\operatorname{sh}\varphi$, $t=\operatorname{ch}\varphi$, $t=\sin\varphi$

☆ 非初等函数

$\Phi(x) = \displaystyle\int e^{-x^2}\,dx$ (欧拉-泊松积分)

$li(x) = \displaystyle\int \frac{1}{\ln x}\,dx$ 积分对数

$si(x) = \displaystyle\int \frac{\sin x}{x}\,dx$ 积分正弦, $ci(x) = \displaystyle\int \frac{\cos(x)}{x}\,dx$

· 椭圆积分 $\displaystyle\int R\left(x, \sqrt{P_n(x)}\right)dx$, $n>2$

$n=3,4$. I: $\displaystyle\int \frac{d\varphi}{\sqrt{1-k^2\sin^2\varphi}}$, II: $\displaystyle\int \sqrt{1-k^2\sin^2\varphi}\,d\varphi$, III: $\displaystyle\int \frac{d\varphi}{(1+h\sin^2\varphi)\sqrt{1-k^2\sin^2\varphi}}$

eg. $\displaystyle\int \frac{dx}{\sqrt{1+x^4}}$

$S(x) = \displaystyle\int \sin(x^2)\,dx$
$C(x) = \displaystyle\int \cos(x^2)\,dx$
} 菲涅尔积分

补充：利用复数计算积分

$\cos x = \mathrm{Re}(e^{ix})$ 即实部

① $\int e^x \sin x\,dx$ ，$e^{ix} = \cos x + i\sin x$ ，$\sin x = \mathrm{Im}(e^{ix})$ 即虚部

$$\mathrm{Im}\left(\int e^x e^{ix}\,dx\right) = \mathrm{Im}\left(\int e^{(1+i)x}\,dx\right) = \mathrm{Im}\left(\frac{1}{1+i}e^{(1+i)x}\right) + C$$

$$\frac{1-i}{2} e^x(\cos x + i\sin x) = \frac{e^x}{2}(\cos x + \sin x) + i\,\frac{e^x}{2}(\sin x - \cos x)$$

从而 $\int e^x \sin(x)\,dx = \frac{e^x}{2}(\sin x - \cos x) + C$ ，同理 $\int e^x \cos x\,dx = \frac{e^x}{2}(\sin x + \cos x) + C$

② $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$ ，$\int \frac{dx}{\sqrt{1+x^2}} \overset{x=iu}{=\!=} \int \frac{i\,du}{\sqrt{1-u^2}} = i\arcsin(-ix) + C = g(x) + C$

注：可看 Page 14

利用 $\sin(ix) = i\,\mathrm{sh}(x)$

记 $i\arcsin(-ix) = g(x)$ ，则 $\sin(-i g(x)) = -ix \Leftrightarrow \sin(i g(x)) = ix \Leftrightarrow g(x) = \mathrm{arsh}(x)$

$\sin(i g(x)) = i\,\mathrm{sh}(g(x))$

由于 $\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

$\frac{e^{g(x)} - e^{-g(x)}}{2i}$ ，从而 $e^{g(x)} - e^{-g(x)} = 2x$

记 $e^{g(x)} = y$ ，则 $y - \frac{1}{y} = 2x$ ，$y^2 - 2xy - 1 = 0$

$y = \frac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$ ，由于 $y = e^{g(x)} > 0$ ，$\pm$ 取 $+$

故 $\int \frac{dx}{\sqrt{1+x^2}} = g(x) + C = \ln y + C = \ln\left(x + \sqrt{x^2+1}\right) + C$

$$\int \frac{dx}{\sqrt{x^2-1}} \overset{x=it}{=\!=\!=} \int \frac{dt}{\sqrt{1+t^2}} = \ln\left(t+\sqrt{t^2+1}\right)+C = \ln\left(-ix+i\sqrt{x^2-1}\right)+C$$

$$= \ln\left|x-\sqrt{x^2-1}\right|+\tilde{C} \quad , \quad \ln(-i) = -\frac{\pi}{2}i$$

§ 4.4 有理函数的积分

① $\int \frac{P(x)}{Q(x)}dx$，$P(x), Q(x)$ 为多项式

多项式
<u>代数基本定理</u>

基础：
① 多项式都可分解 $P_n(x) = A\prod_{i=1}^{n}(x-x_i) = A(x-x_1)(x-x_2)\cdots(x-x_n)$

x_i 不一定是实数

② 也都可分解 $P_n(x) = A\prod_{i=1}^{m}(x-x_i)^{\alpha_i}\prod_{j=1}^{k}(x^2+b_jx+c_j)^{\beta_j}$

x_i：实数；$x^2+b_jx+c_j$：没有实根

$x_i = \bar{x}_j$

$(x-x_i)(x-\bar{x}_i)$

然后：$\frac{P(x)}{Q(x)}$ 总可分解成如下形式的和之后

$$\frac{A_1}{x-a} + \cdots + \frac{A_k}{(x-a)^k}$$

$$\frac{B_1x+C_1}{x^2+px+q} + \cdots + \frac{B_kx+C_k}{(x^2+px+q)^k}$$

x^2+px+q 没有实根

$t^2+r^2 = \left(\left(\boxed{\frac{t}{r}}\right)^2+1\right)r^2$

看成整体 $\tilde{t}$

从而只需求 $\int\frac{dx}{(x-a)^k}$，$\int\frac{x\,dx}{(x^2+px+q)^k}$ 和 $\int\frac{dx}{(x^2+px+q)^k}$

$k=1$. $\ln|x-a|+C$

$k\neq 2$ $\frac{1}{1-k}(x-a)^{1-k}+C$

$\int\frac{t\,dt}{(\boxed{t^2+1})^k}$ 和 $\int\frac{dt}{(t^2+1)^k}$

$t\,dt = \frac{1}{2}d(t^2+1)$

$$\int \frac{\overset{1+t^2-t^2}{1}\,dt}{(t^2+1)^k} = I_k$$

$$= \underbrace{\int \frac{dt}{(1+t^2)^{k-1}}}_{I_{k-1}} - \underbrace{\int \frac{t\cdot t}{(1+t^2)^k}\,dt}_{=\frac{1}{2}\frac{1}{k-1}\left(\frac{t}{(t^2+1)^{k-1}} - I_{k-1}\right)}$$

$$\frac{t}{(1+t^2)^k}\,dt = \frac{1}{2}\,\frac{-1}{k-1}\,d\left(\frac{1}{(1+t^2)^{k-1}}\right)$$

$$= \frac{1}{2(k-1)}\,\frac{t}{(t^2+1)^{k-1}} + \frac{2k-3}{2(k-1)}\,I_{k-1}$$

可利用递推得到 I_k 的结果.

$k=1$, $= \frac{1}{2}\ln(t^2+1)+C$

$k\neq 1$, $\frac{1}{2}\frac{1}{1-k}(t^2+1)^{1-k}+C$

$\boxed{\text{结论}}$ 有理函数的不定积分
计算出来的
都是初等函数

代数

练习: $\frac{7}{18} = \frac{?}{2} + \frac{?}{3} + \frac{?}{9} = \frac{1}{2} - \frac{1}{9}$

$\frac{17}{18} = \frac{?}{2} + \frac{?}{3} + \frac{?}{9} = \frac{1}{2} + \frac{1}{3} + \frac{1}{9}$

$18 = 2\times3\times3 = 2\times3^2$

$\frac{A}{2} + \frac{B}{3^2} = \frac{17}{18} = \frac{17}{2\times3^2}$

$3^2A + 2B = 17$

互素/互质, 最大公约数=1

② 化为有理函数

$\int R(\sin(x), \cos(x))\,dx$ ， $\int R\left(x, \sqrt[n]{\frac{ax+b}{cx+d}}\right)dx$ ， $\int R\left(x, \sqrt{ax^2+bx+c}\right)dx$

$t = \tan\frac{x}{2}$ ， $dx = \frac{2dt}{1+t^2}$

$\sin x = \frac{2t}{1+t^2}$

$\cos x = \frac{1-t^2}{1+t^2}$

万能公式

t

$x = \frac{t^n d - b}{a - ct^n}$

$a=0$ 时

配方

$1+t^2$, t^2-1, $1-t^2$ （$a \neq 0$ 时）

$t = \mathrm{sh}\varphi$, $t = \mathrm{ch}\varphi$, $t = \sin\varphi$

例： $\int \frac{1+\sin(x)}{\sin(x)\left(1+\cos(x)\right)}\,dx$ ， $\int \frac{\sqrt{x-1}}{x}\,dx$ ， $\int \frac{dx}{\left(1+\sqrt[3]{x}\right)\sqrt{x}}$

$t = \tan\frac{x}{2}$ （例 4）

$t = \sqrt{x-1}$ （例 5）

$t = \sqrt[6]{x}$ （例 7）

例. $\int \frac{x+1}{x^2-5x+6}\,dx$

（例 1）

§ 4.5 积分表的使用

→ 先把问题简化成好查表的情形

线性，变换，分部积分

Page 374: 附录四

补记：不定积分的计算技巧(II)

$\sin^2 x + \cos^2 x = 1$

书上细则

$\int \sin^{2k+1}(x)\cos^n(x)\,dx$, $u = \cos x$

$\int \sin^n(x)\cos^{2k+1}(x)\,dx$, $u = \sin x$

组成 u 的函数

eg $\int \frac{\sin x}{\cos x}\,dx = -\int \frac{d\cos x}{\cos x} = -\ln|\cos x| + C$

$\int \sin^{2k}(x)\cos^{2l}(x)\,dx$ ，利用 $\sin^2(x) = \frac{1-\cos(2x)}{2}$ ， $\cos^2(x) = \frac{1+\cos(2x)}{2}$

→ 化成 $\cos(2x)$ 的多项式

$\int \tan^n(x)\sec^{2k}(x)\,dx$ ， $\int \tan^{2k-1}(x)\sec^n(x)\,dx$ ，

$u = \tan x$ $\quad\quad$ $u = \sec x$

$$\int \sin^n(x)\,dx = -\sin^{n-1}(x)\cos x + \int \underbrace{\cos^2 x}_{1-\sin^2 x} \cdot (n-1)\sin^{n-2}x\,dx$$

$-\sin^{n-1}(x)\,d\cos x$

$\Downarrow$

$$\int \sin^n(x)\,dx = -\frac{\sin^{n-1}(x)\cos x}{n} + \frac{n-1}{n}\int \sin^{n-2}(x)\,dx$$

$\int \sin^m(x)\cos^n(x)\,dx$. Page 380. 公式 55. 两种递推公式

$$\int \sin^n x\,dx = -\frac{\sin^{n-1}x\cos x}{n} + \frac{n-1}{n}\int \sin^{n-2}x\,dx. \rightarrow$$

$\int \sin x\,dx$

$\int 1\,dx$

综合练习.

$$\int x\cos x\,dx = \int x\,d\sin x = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$$

$$\int x^3 \ln x\,dx = \int \ln x\,d\left(\frac{x^4}{4}\right) = \frac{x^4}{4}\ln x - \int \frac{x^4}{4}\cdot\frac{1}{x}\,dx = \frac{x^4}{4}\ln x - \frac{x^4}{16} + C$$

$$\int \frac{dx}{x^2\sqrt{x^2-1}}$$

① $x = \sec t = \dfrac{1}{\cos t}$

$$\int \frac{\sec t\cdot\tan t}{\sec^2 t\cdot\tan t}\,dt = \int \cos t\,dt = \sin t + C = \frac{1}{x}\sqrt{x^2-1} + C.$$

|| ② $\dfrac{1}{x} = t$

$$\int \frac{dx}{x^3\sqrt{1-\frac{1}{x^2}}} = \int \frac{-1}{x}\frac{1}{\sqrt{1-\left(\frac{1}{x}\right)^2}}\,d\left(\frac{1}{x}\right) = \int \frac{\overbrace{-t\,dt}^{\frac{1}{2}d(1-t^2)}}{\sqrt{1-t^2}} = \sqrt{1-t^2} + C = \frac{1}{x}\sqrt{x^2-1} + C.$$

$$\int \frac{dx}{a^2\sin^2 x + b^2\cos^2 x},\quad ab \neq 0$$

① 万能公式，$t = \tan\frac{x}{2}$

② $t = \tan x$

||

$$\int \frac{dt}{a^2t^2 + b^2} = \frac{1}{ab}\arctan\left(\frac{at}{b}\right) + C = \frac{1}{ab}\arctan\left(\frac{a}{b}\tan x\right) + C$$